WSTP F_U_Bi_i Symbolic Export — Wolfram Language Expression #51

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PAPER_987: WSTP F_U_Bi_i Symbolic Export — Wolfram Language Expression #51

Abstract

Expression #51 in `wstp_kernel_demo_runner.py` encodes the complete 6-layer F_{U,Bi_i} master equation in Wolfram Language for symbolic evaluation via the Wolfram Symbolic Transfer Protocol (WSTP). The export enables closed-form simplification, series expansion, and parametric analysis of all six layers simultaneously within the Mathematica kernel.

1. Wolfram Language Definition

```
Module[{G, Msun, Rsun, AU, SSq, betai, kappa, omegaSCm, ...},
  (* Constants *)
  G = 6.674*^-11; Msun = 1.989*^30; ...

  (* NOTE: G*M/r^2 in layer functions = Step 10 Newton observational projection form; canonica
  (* Layer functions *)
  Ug26[M_, r_] := Sum[G*M/r^2 * SSq * i/26, {i, 1, 26}];
  Ub26[M_, r_] := Sum[G*M/r^2 * Exp[-SSq*i/26] * betai, {i, 1, 26}];
  Um[M_, r_]  := G*M/r^2 * 1*^-8;
  UA[M_, r_]  := -G*M/r^2 * 1*^-10;
  Fn[M_, r_]  := G*M/r^2 * 1*^-7;
  Phi[gamma_] := Exp[-(omegaSCm - omegaSCm)^2/(2*gamma^2)] * S26;
  Enet[t_, gamma_] := (2*(Ub26[M,r]/(Ug26[M,r]+0.001))-1)*Exp[kappa*t]*S26;

  (* Master equation *)
  FUBii[M_, r_, t_, gamma_] :=
    Ug26[M,r] + Um[M,r] + UA[M,r] - Ub26[M,r] +
    Fn[M,r] * S26 * Phi[gamma] * Enet[t,gamma];

  (* Evaluate at three radii *)
  Table[FUBii[Msun, R, 86400, 0.1*^12], {R, {Rsun, AU, 10*AU}}]
]
```

2. Symbolic Capabilities

The WSTP export enables: - **Series expansion:** `Series[FUBii[M, r, t, g], {r, Infinity, 3}]` — asymptotic behavior - **Differentiation:** `D[FUBii[M, r, t, g], r]` — force gradient (tidal) - **Integration:** `Integrate[FUBii[M, r, t, g], {r, Rsun, AU}]` — work done - **Simplification:** `FullSimplify[FUBii[M, r, t, g]]` — closed-form reduction - **Parametric plots:** `Plot3D[FUBii[Msun, r, t, 0.1*^12], {r, 1*^8, 1*^12}, {t, 0, 86400}]`

3. Integration with WSTP Pipeline

Expression #51 joins the existing 50 WSTP expressions. The WSTP kernel demo runner dispatches it via:

```
link.putFunction("EvaluatePacket", 1)
link.put(expr_51_code)
link.endPacket()
result = link.getResult()
```

4. Implementation

Added as expression #51 in `wstp_kernel_demo_runner.py`, function `_build_expressions()`. Total WSTP expressions: 51.

References

- PAPER_979: Complete 6-Layer F_U_Bi_i
- PAPER_985: Production Kernel

Session 225 Phonon-Physics Upgrade: S ³ Ramanujan Summation

> *Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1)\cdots(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\mathrm{SSq}]| < 1$ (satisfied by $[\mathrm{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\mathrm{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4}\,\text{day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25$ THz	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37}\,\text{kg/m}^3$	Fundamental

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency ω_{SCm}	$2\pi \times 1.25$ THz (Pd-D lattice)	Measured Pd-D phonon spectrum	Fukai (2005)	Mapped to SCm
Vacuum energy ρ_{vac}	7.09×10^{-37} kg/m^3	$\rho_{\text{vac}} \sim 10^{-29}$ g/cm^3	Planck 2018	Novel SCm scale
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator).

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§A. Cosmogenesis-Linked Lagrangian

The Wolfram export allows symbolic computation of $\delta S/\delta\phi$ directly in Mathematica, cross-validating the numerical Euler-Lagrange result from PAPER_981.

§B. VDS/DVP/BSH Deep Synthesis

- **VDS:** Wolfram's `Sum` and `Exp` handle the vacuum density series analytically.
- **DVP:** Symbolic differentiation reveals the DPM dipole contribution at each order.
- **BSH:** The buoyancy harmonic sum `Sum[Exp[-SSq*i/26]*betai, {i,1,26}]` evaluates to closed form via geometric series.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).*

Paper	Title
PAPER_1068	Wolfram Physics Bridge WSTP Symbolic Export
PAPER_1078	QCalcGeom Master Equation Derivation

2 cross-reference(s) identified.